DAA LAB ASSIGNMENT WEEK-2

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1)Bubble Sort

CODE:

```
#include <stdio.h>

void BubbleSort(int arr[], int size) {
 int swapped;
 do {
 swapped = 0;
 for (int i = 0; i < size - 1; i++) {
 if (arr[i] > arr[i + 1]) {
 int temp = arr[i];
 arr[i] = arr[i + 1];
 arr[i + 1] = temp;
 swapped = 1;
 }
 }
 } while (swapped);
}

void Display(int arr[], int size) {
 for (int i = 0; i < size; i++) {
```

```
 printf("%d ", arr[i]);
 }
 printf("\n");
}

int main() {
 int n;
 printf("Enter the size of the array: ");
 scanf("%d", &n);

 int arr[n];
 printf("Enter the values in the array:\n");

 for (int i = 0; i < n; i++) {
 printf("%d. ", i + 1);
 scanf("%d", &arr[i]);
 }

 printf("The array is: ");
 Display(arr, n);
 BubbleSort(arr, n);
 printf("The array after sorting is: ");
 Display(arr, n);
 return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./bubblesort
Enter the size of the array: 01
Enter the values in the array:
1. 5
The array is: 5
The array after sorting is: 5
santhan@Santhan-PC:~$ ./bubblesort
Enter the size of the array: 10
Enter the values in the array:
1. 67
2. 897
3. 546
4. 467
5. 923
6. 234
7. 523
8. 4
9. 679
10. 56
The array is: 67 897 546 467 923 234 523 4 679 56
The array after sorting is: 4 56 67 234 467 523 546 679 897 923
```

2)Selection Sort

CODE:

```
#include <stdio.h>


void SelectionSort(int arr[], int size){
 for(int i=0; i<size; i++){
 int smallest = arr[i];
 for(int j=i; j<size; j++){
 if(arr[j]<arr[i]){
 int temp = arr[i];
 arr[i] = arr[j];
 arr[j] = temp;
 }
```

```
 }
 }
}

void Display(int arr[], int size) {
 for (int i = 0; i < size; i++) {
 printf("%d ", arr[i]);
 }
 printf("\n");
}

int main() {
 int n;
 printf("Enter the size of the array: ");
 scanf("%d", &n);

 int arr[n];
 printf("Enter the values in the array:\n");

 for (int i = 0; i < n; i++) {
 printf("%d. ", i + 1);
 scanf("%d", &arr[i]);
 }

 printf("The array is: ");
 Display(arr, n);
```

```
 SelectionSort(arr, n);


 printf("The array after sorting is: ");

 Display(arr, n);


 return 0;

}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./selectionsort
Enter the size of the array: 10
Enter the values in the array:
1. 5
2. 24
3. 9876
4. 4578
5. 456
6. 23
7. 890
8. 589
9. 2309
10. 2106
The array is: 5 24 9876 4578 456 23 890 589 2309 2106
The array after sorting is: 5 23 24 456 589 890 2106 2309 4578 9876
```

3)Insertion Sort

CODE:

```
#include <stdio.h>


void InsertionSort(int arr[], int size) {

 for (int i = 1; i < size; ++i) {

 int key = arr[i];

 int j = i - 1;
```

```
 while (j >= 0 && arr[j] > key) {
 arr[j + 1] = arr[j];
 j--;
 }
 arr[j + 1] = key;
 }
}

void Display(int arr[], int size) {
 for (int i = 0; i < size; i++) {
 printf("%d ", arr[i]);
 }
 printf("\n");
}

int main() {
 int n;
 printf("Enter the size of the array: ");
 scanf("%d", &n);

 int arr[n];
 printf("Enter the values in the array:\n");

 for (int i = 0; i < n; i++) {
 printf("%d. ", i + 1);
 scanf("%d", &arr[i]);
```

```
 }

 printf("The array is: ");
 Display(arr, n);
 InsertionSort(arr, n);
 printf("The array after sorting is: ");
 Display(arr, n);
 return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./insertionsort
Enter the size of the array: 10
Enter the values in the array:
1. 345
2. 697
3. 895
4. 34
5. 67
6. 21
7. 4567
8. 8902
9. 4689
10. 498
The array is: 345 697 895 34 67 21 4567 8902 4689 498
The array after sorting is: 21 34 67 345 498 697 895 4567 4689 8902
```

4)Heap Sort

CODE:

```
#include <stdio.h>

void heapify(int arr[], int n, int i)
{
 int largest = i;
```

```
 int l = 2 * i + 1;
 int r = 2 * i + 2;
 if (l < n && arr[l] > arr[largest])
 largest = l;
 if (r < n && arr[r] > arr[largest])
 largest = r;
 if (largest != i)
 {
 int temp = arr[i];
 arr[i] = arr[largest];
 arr[largest] = temp;

 heapify(arr, n, largest);
 }
}

void heapSort(int arr[], int n)
{
 for (int i = n / 2 - 1; i >= 0; i--)
 heapify(arr, n, i);
 for (int i = n - 1; i > 0; i--)
 {
 int temp = arr[0];
 arr[0] = arr[i];
 arr[i] = temp;
 heapify(arr, i, 0);
```

```
 }
}

void Display(int arr[], int size) {
 for (int i = 0; i < size; i++) {
 printf("%d ", arr[i]);
 }
 printf("\n");
}

int main()
{
 int n;
 printf("Enter the size of the array: ");
 scanf("%d", &n);

 int arr[n];
 printf("Enter the values in the array:\n");

 for (int i = 0; i < n; i++) {
 printf("%d. ", i + 1);
 scanf("%d", &arr[i]);
 }

 printf("The array is: ");
 Display(arr, n);
```

```
 heapSort(arr, n);


 printf("The array after sorting is: ");
 Display(arr, n);


 return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./heapsort
Enter the size of the array: 6
Enter the values in the array:
1. 34
2. 78
3. 0
4. 3
5. 1
6. 58
The array is: 34 78 0 3 1 58
The array after sorting is: 0 1 3 34 58 78
```

5)Bucket Sort

CODE:

```
#include <stdio.h>
#include <stdlib.h>


struct Node{
 int data;
```

```
 struct Node *next;
};

struct Node* insertSorted(struct Node *head, int value);

void bucketSort(int arr[], int n, int bucketCount);

void Display(int arr[], int n);

struct Node* insertSorted(struct Node *head, int value) {
 struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
 newNode->data = value;
 newNode->next = NULL;
 if (head == NULL || value < head->data) {
 newNode->next = head;
 return newNode;
 }
 struct Node *current = head;
 while (current->next != NULL && current->next->data < value) {
 current = current->next;
 }
 newNode->next = current->next;
 current->next = newNode;
 return head;
}
```

```
void bucketSort(int arr[], int n, int bucketCount) {
 int i, j = 0;
 int max = arr[0];
 for (i = 1; i < n; i++) {
 if (arr[i] > max)
 max = arr[i];
 }

 struct Node *buckets[bucketCount];
 for (i = 0; i < bucketCount; i++) {
 buckets[i] = NULL;
 }

 for (i = 0; i < n; i++) {
 int bucketIndex = (arr[i] * bucketCount) / (max + 1);
 buckets[bucketIndex] = insertSorted(buckets[bucketIndex],arr[i]);
 }

 for (i = 0; i < bucketCount; i++) {
 struct Node *current = buckets[i];
 while (current != NULL) {
 arr[j++] = current->data;
 current = current->next;
 }
 }
}
```

```
void Display(int arr[], int n) {
 for (int i = 0; i < n; i++)
 printf("%d ", arr[i]);
 printf("\n");
}

int main() {
 int n;
 int bucketCount = 5;
 printf("Enter the size of the array: ");
 scanf("%d", &n);

 int arr[n];
 printf("Enter the values in the array:\n");

 for (int i = 0; i < n; i++) {
 printf("%d. ", i + 1);
 scanf("%d", &arr[i]);
 }

 printf("The array is: ");
 Display(arr, n);
 bucketSort(arr, n, bucketCount);
 printf("The array after sorting is: ");
 Display(arr, n);
```

```
 return 0;

}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./bucketsort
Enter the size of the array: 10
Enter the values in the array:
1. 4
2. 34
3. 67
4. 89
5. 45
6. 23
7. 12
8. 37
9. 82
10. 91
The array is: 4 34 67 89 45 23 12 37 82 91
The array after sorting is: 4 12 23 34 37 45 67 82 89 91
```